

National University of Computer and Emerging Sciences, Lahore Campus

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	Practice Problem:	Dimensional Modeling - Solution

Consider the following three dimensions and a base fact table:

Semester (SemID, SemDescription, AcademicYearID, AcademicYearDescription)
Course (CourseCode, CourseDescription, OfferingSchoolID, OfferingSchoolDescription)
Student (StudentID, StudentDescription, BatchID, BatchDescription)
Registration (SemID, CourseID, StudentID, GPA, LetterGrade, RegistrationCount (always=1))

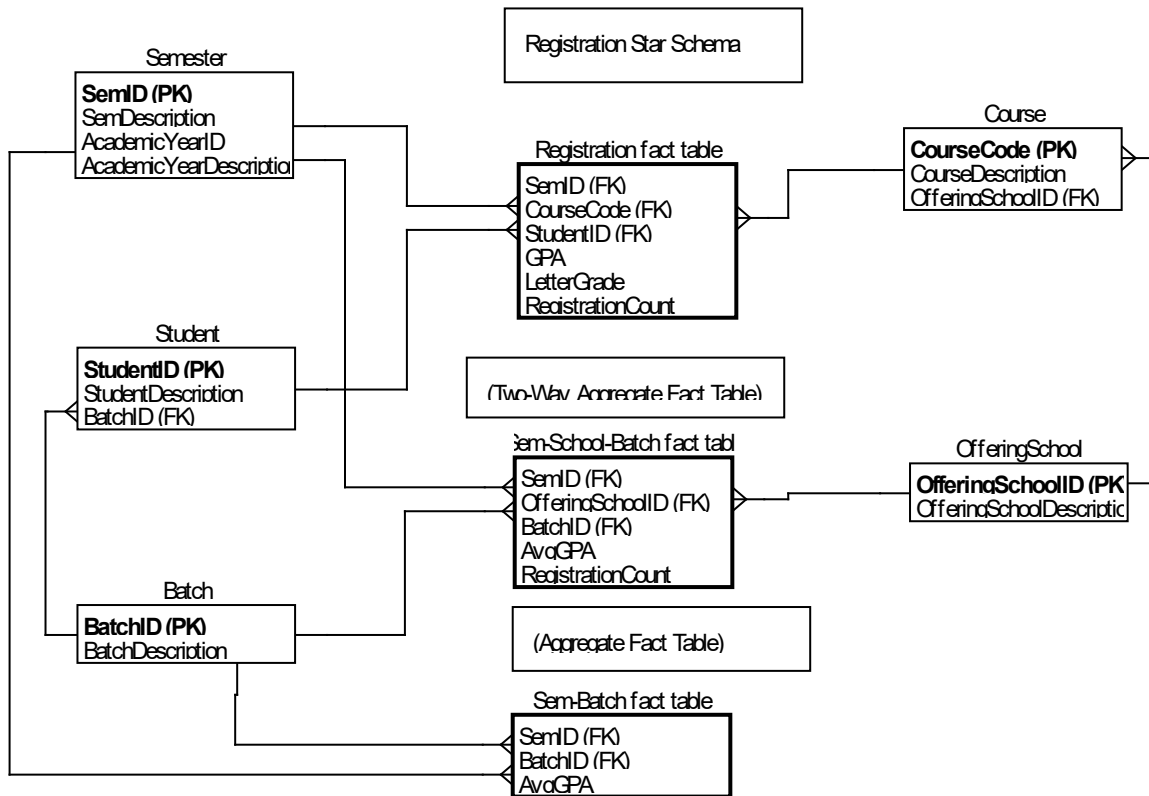
Following queries are made most frequently:

Query 1. Average GPA and total number of registered students by semester by offering school by batch.
Query 2. Average GPA by semester by batch.

Assume: 12 semesters, 6 academic years, 400 courses, 10 schools, 5000 students, and 5 batches.

Q1. Draw a star schema that includes registration base fact table and aggregate fact tables for the above requirements. Take appropriate assumption, if required. Show the primary keys, foreign keys and all the relationships between the dimensions and fact tables. Note: Draw only a single diagram that includes base fact table as well as aggregate fact tables.

Ans:



Q2. Estimate the size (in number of rows) of the above base fact table, aggregate fact tables, semester dimension, and student dimension.

Ans:

Base Fact Table: $12 \times 400 \times 5000 = 24,000,000$ rows

Aggregate Fact Table1: 12 (semester) $\times 10$ (offering schools) $\times 5$ (batch) = 600 rows

Aggregate Fact Table2: 12 (semester) $\times 5$ (batch) = 60 rows

Semester dimension: 12 rows

Student dimension: $5000 + 5 = 5005$ rows

Q3. Suppose there are 5 brands and 200 products in each brand. There are 3 countries and 5 store locations in each country. Also assume there is at least one sale per product per store per week in the sales base fact table. Estimate the number of rows of fact table retrieved and summarized for following types of queries:

	Product	Store	Time	# of Rows retrieved
Query 1:	1 product	1 store	1 week	$1 \times 1 \times 1 = 1$
Query 2:	1 product	All stores	1 week	$1 \times 15 \times 1 = 15$
Query 3:	1 brand	1 store	1 week	$200 \times 1 \times 1 = 200$
Query 4:	1 brand	1 country	1 year	$200 \times 5 \times 52 = 52,000$
Query 5:	All brands	All stores	1 year	$1000 \times 15 \times 52 = 780,000$

Q4. For which of the above queries Aggregate fact tables should be used and why?

Ans: For Queries 3, 4, and 5.

Q5. Suppose you created an aggregate fact table for Query 3, then how many rows do you need to retrieve for Queries 3, 4 and 5?

Query 3:	1 brand	1 store	1 week	$1 \times 1 \times 1 = 1$
Query 4:	1 brand	1 country	1 year	$1 \times 5 \times 52 = 260$
Query 5:	All brands	All stores	1 year	$5 \times 15 \times 52 = 3900$